

# EECS 841: Computer Vision

## Lecture 9

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# Outline: Part III

## Binary Image Processing:

- Morphological Operations

## Binary/Color Image Processing for Segmentation

- Morphological Operations
- Labeling Regions
- Computing Distance in a Digital Image
- Region Properties
- Boundary Representations

→ Separating a “common” region for labelling followed by analyzing → ex: lung segmentation in biomedical domain

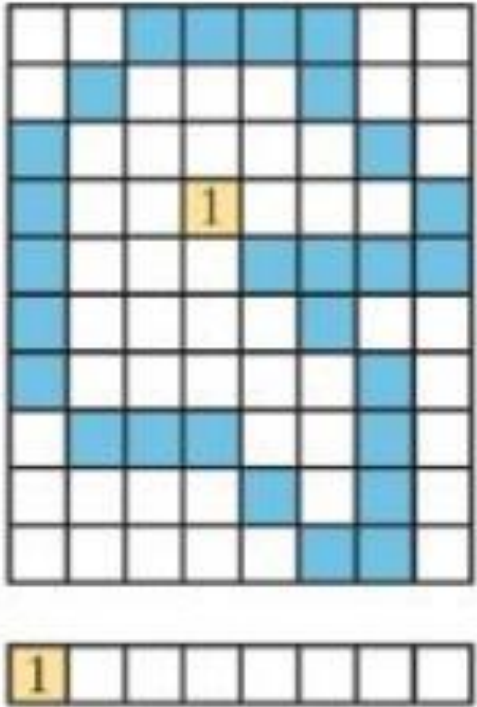
# Application of “Adjacency”

How is 4/8 adjacency helpful:

- Helps to show connectivity → connectivity is the basis of different application
  - connected pixels be part of an object – helpful for object segmentation
  - region coloring by 4/8-adjacency you can perform coloring too, ex. flood-fill algorithm
  - region growing can also be conducted with 4/8 adjacency

# Flood-fill Algorithm

- **Floodfill (seed fill):** the problem of coloring all the pixels that are connected to a *seed pixel* with some desired new color.



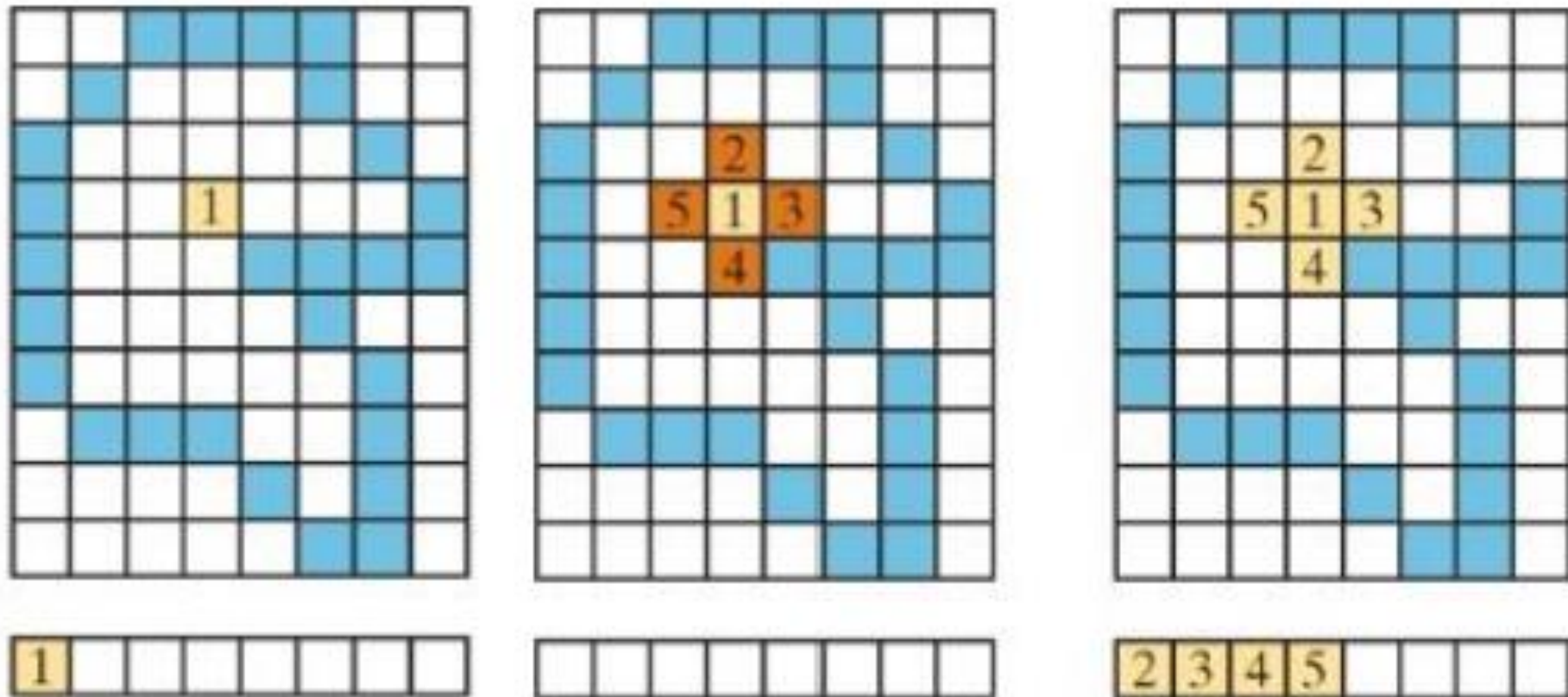
Need to know them to do flood filling:

- Boundary pixel (either by location or intensity)
- By what adjacency method the region will grow
- Starting seed
- Is seed going to be moving
- What color you want to fill in with

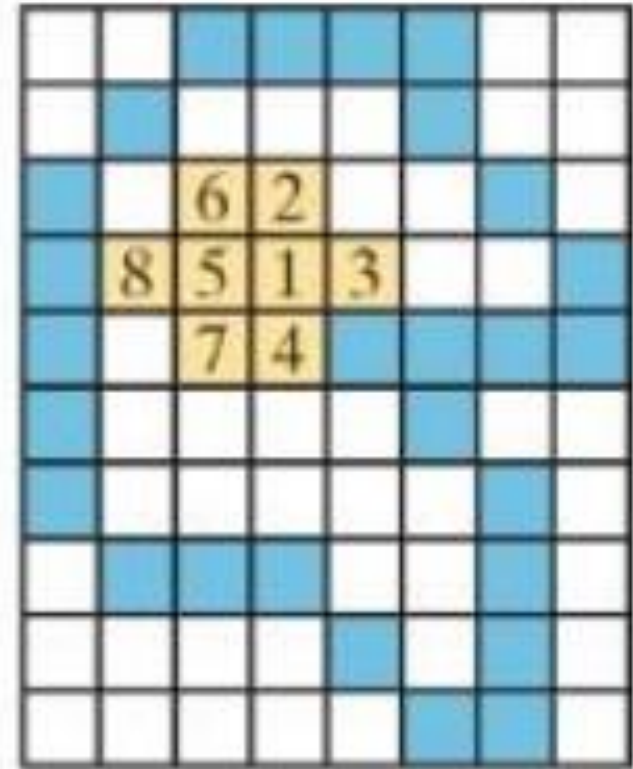
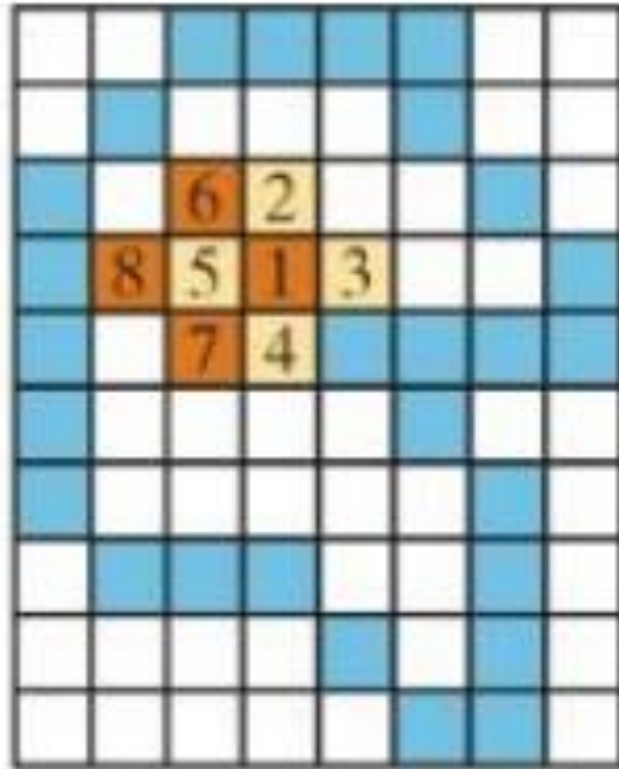
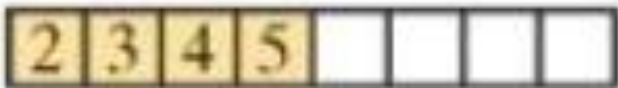
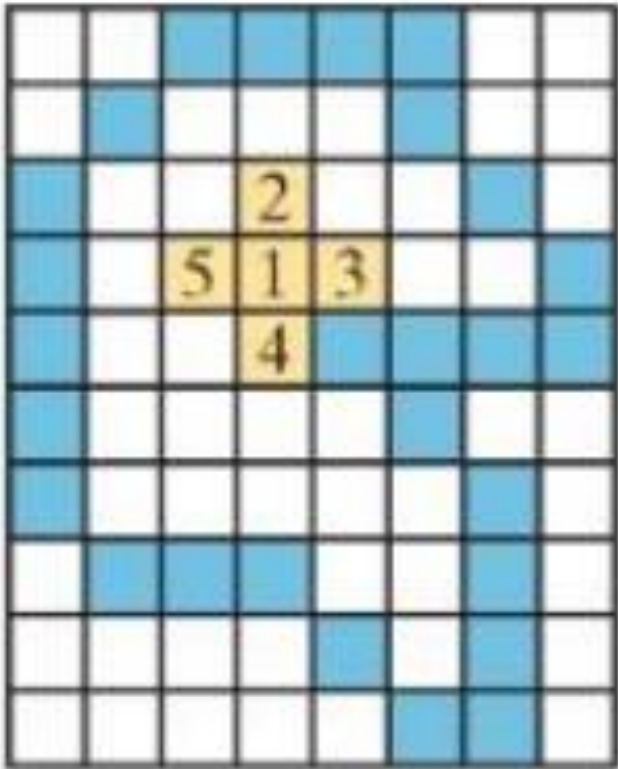
Ex. Perform flood fill on the left image where blue pixels are boundary, and you are to assign a new color (e.g., yellow) only the pixels inside the blue boundary. You may start from the seed pixel labelled as “1”. Consider 4 adjacency neighbor to grow your region and color until completed. Your seed value may update when necessary.

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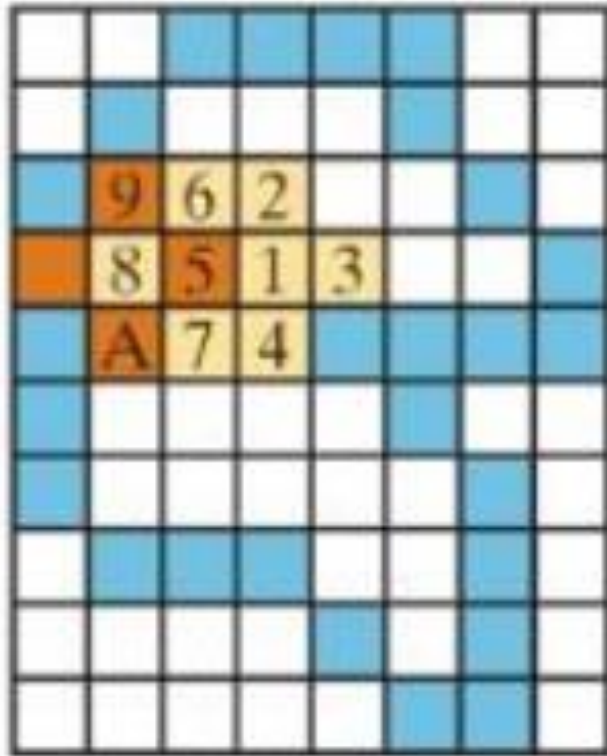


# Flood-fill Algorithm

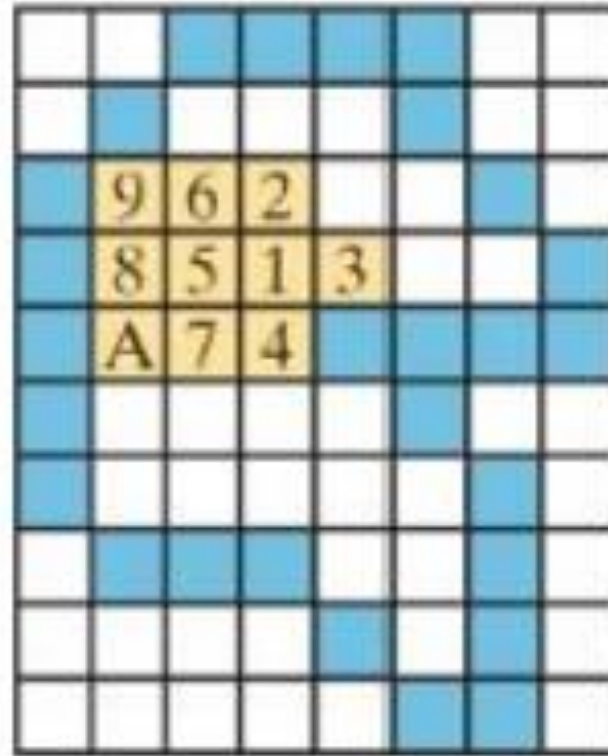




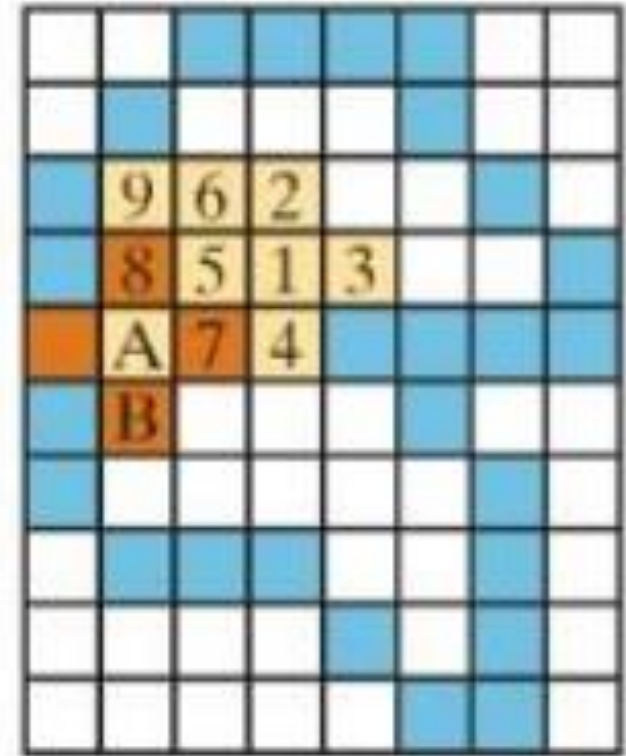
# Flood-fill Algorithm



|   |   |   |   |   |  |  |  |
|---|---|---|---|---|--|--|--|
| 2 | 3 | 4 | 6 | 7 |  |  |  |
|---|---|---|---|---|--|--|--|

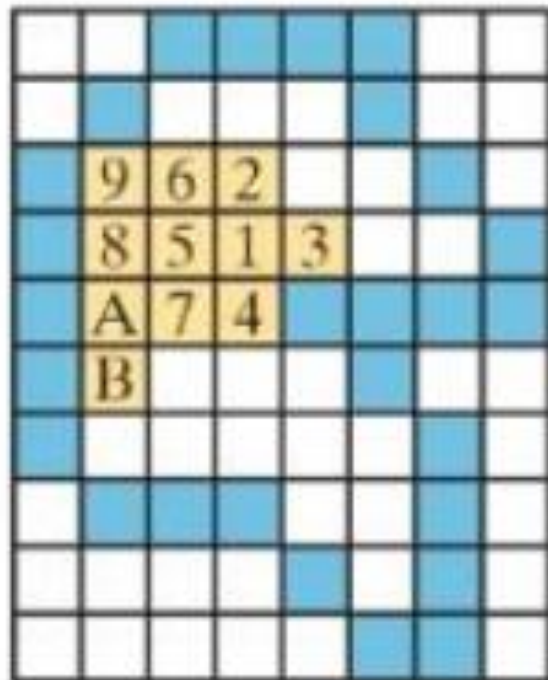


|   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|--|
| 2 | 3 | 4 | 6 | 7 | 9 | A |  |
|---|---|---|---|---|---|---|--|

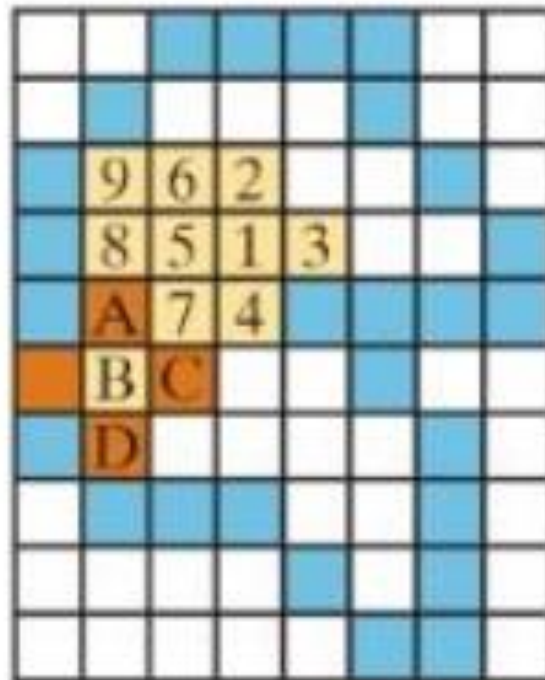


|   |   |   |   |   |   |  |  |
|---|---|---|---|---|---|--|--|
| 2 | 3 | 4 | 6 | 7 | 9 |  |  |
|---|---|---|---|---|---|--|--|

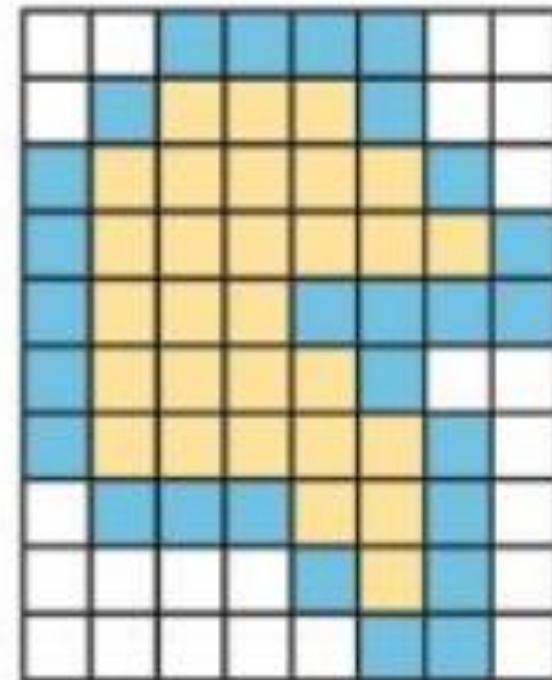
# Flood-fill Algorithm



|   |   |   |   |   |   |   |  |
|---|---|---|---|---|---|---|--|
| 2 | 3 | 4 | 6 | 7 | 9 | B |  |
|---|---|---|---|---|---|---|--|



|   |   |   |   |   |   |  |  |
|---|---|---|---|---|---|--|--|
| 2 | 3 | 4 | 6 | 7 | 9 |  |  |
|---|---|---|---|---|---|--|--|

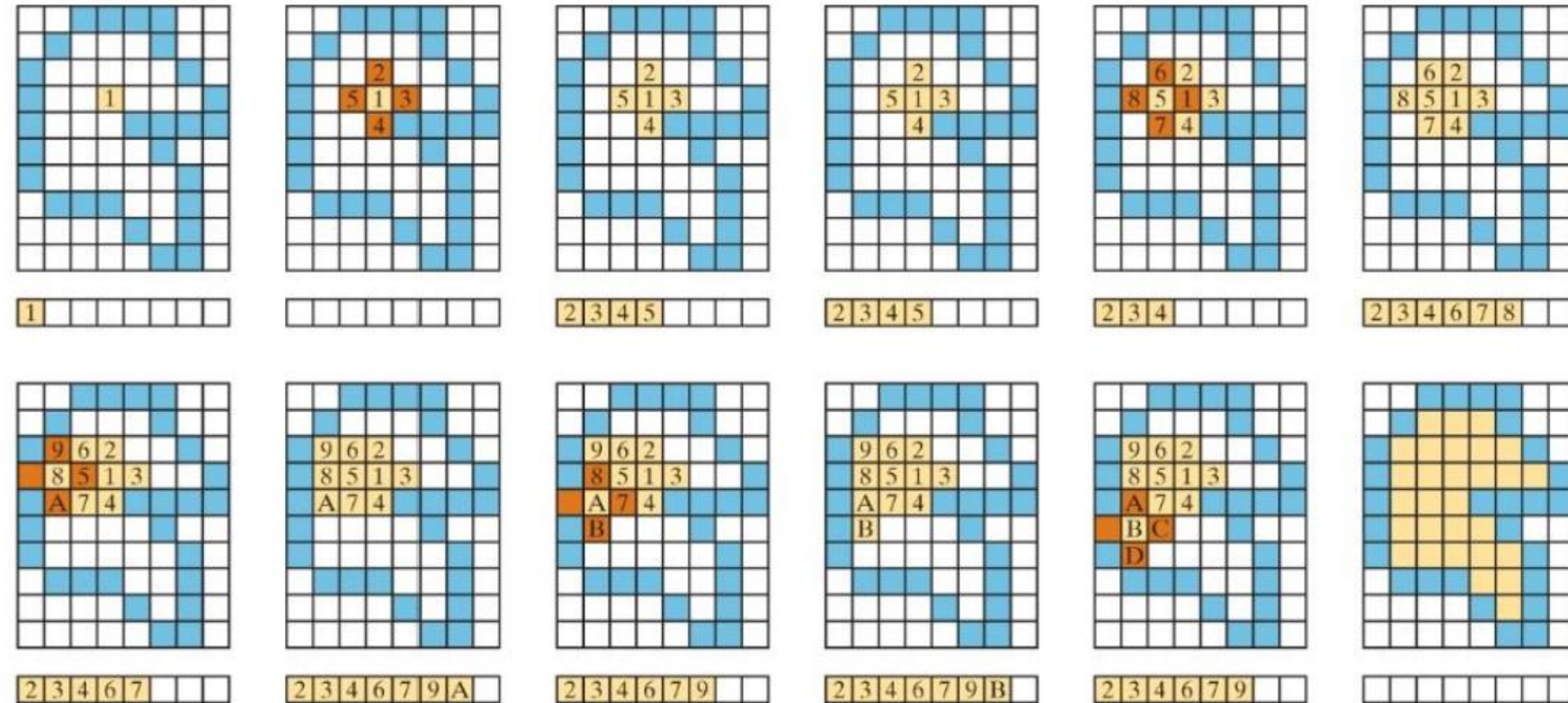




# Flood-fill Algorithm: at a glance

- **Floodfill (seed fill):** the problem of coloring all the pixels that are connected to a *seed pixel* with some desired new color.

**Figure 4.22** Step-by-step illustration of the 4-neighbor FLOODFILL algorithm on a small image. The frontier is shown below the image. Starting from the seed pixel labeled 1, the interior region of white pixels is changed to yellow by the algorithm, while orange is used to indicate the pixels being considered in the current expansion. The labels are artificially introduced to aid in associating pixels in the image with those in the frontier.



# Region Growing

Based on 4-adjacency, apply region growing on the following image considering:

(i) initial label at (2,2), (ii) color = green, and (ii) boundary (=stopping) criteria: region growing for a single iteration as long as difference between the current reference and neighbor maintains a pixel intensity distance  $\leq 2$

|         | Col = 0 | Col = 1 | Col = 2 | Col = 3 |
|---------|---------|---------|---------|---------|
| row = 0 | 0       | 1       | 2       | 0       |
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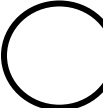
| Checking the given condition | Region should grow? |
|------------------------------|---------------------|
| $7 - 6 = 1$                  | yes                 |
| $7 - 4 = 3$                  | no                  |
| $7 - 5 = 2$                  | yes                 |
| $7 - 3 = 4$                  | no                  |

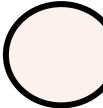
|       |  |  |     |
|-------|--|--|-----|
| (2,2) |  |  | ... |
|-------|--|--|-----|

# Region Growing

Based on 4-adjacency, apply region growing on the following image considering:  
(i) initial point at (2,2) and (ii) threshold  $\leq 2$

Solution:

 = falls in 4 adjacency neighborhood

 = already selected candidate for

 = already colored

 = current reference

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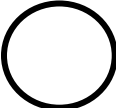
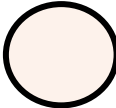
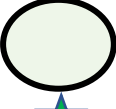
|       |  |  |     |
|-------|--|--|-----|
| (2,2) |  |  | ... |
|-------|--|--|-----|

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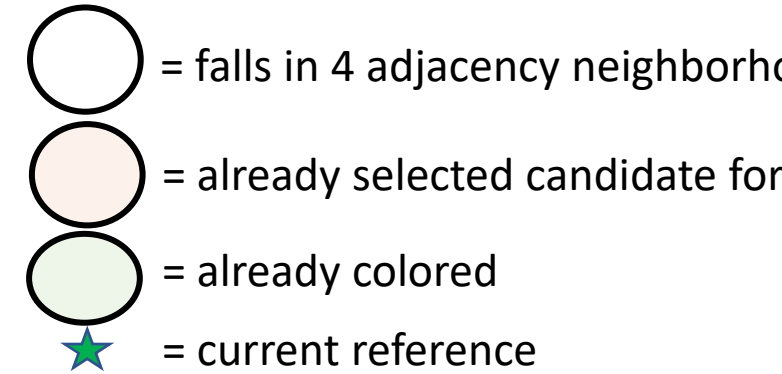
|             |             |     |     |
|-------------|-------------|-----|-----|
| (2,3) = "5" | (2,1) = "6" | ... | ... |
|-------------|-------------|-----|-----|



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(i) initial point at (2,2) and (ii) threshold  $\leq 2$

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| row = 2 | 1       | 4       | 7       | 3       |
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After applying  
threshold

|         | Col = 0 | Col = 1 | Col = 2 | Col = 3 |
|---------|---------|---------|---------|---------|
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|             |              |  |     |
|-------------|--------------|--|-----|
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|-------------|--------------|--|-----|

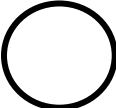
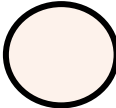
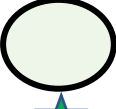
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|       |       |  |  |     |     |
|-------|-------|--|--|-----|-----|
| (2,3) | (1,1) |  |  | ... | ... |
|-------|-------|--|--|-----|-----|

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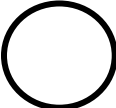
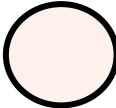
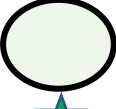
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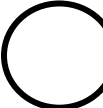
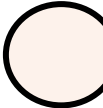
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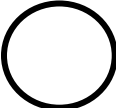
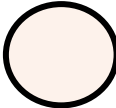
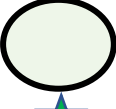
|       |       |  |  |     |     |
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|       |        |  |  |     |     |
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| (2,3) | (1,2)* |  |  | ... | ... |
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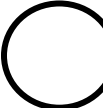
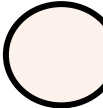
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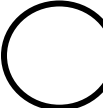
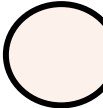
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|-------|--|--|--|-----|-----|

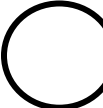
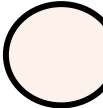
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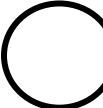
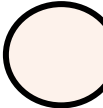
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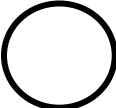
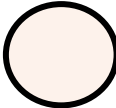
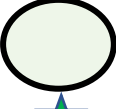
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-  = already colored
-  = current reference

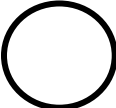
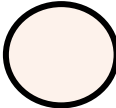
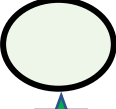
# Region Growing

Based on 4-adjacency, apply region growing on the following image considering:  
(i) initial point at (2,2) and (ii) threshold  $\leq 2$

Solution:

|         | Col = 0 | Col = 1 | Col = 2 | Col = 3 |
|---------|---------|---------|---------|---------|
| row = 0 | 0       | 1       | 2       | 0       |
| row = 1 | 2       | 5       | 6       | 1       |
| row = 2 | 1       | 4       | 7       | 3       |
| row = 3 | 10      | 2       | 5       | 1       |

|        |  |  |  |     |     |
|--------|--|--|--|-----|-----|
| (2,3)* |  |  |  | ... | ... |
|--------|--|--|--|-----|-----|

-  = falls in 4 adjacency neighborhood
-  = already selected candidate for
-  = already colored
-  = current reference

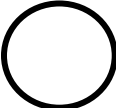
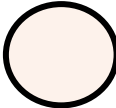
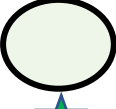
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| row = 1 | 2       | 5       | 6       | 1       |
| row = 2 | 1       | 4       | 7       | 3       |
| row = 3 | 10      | 2       | 5       | 1       |

(Final Ans)

-  = falls in 4 adjacency neighborhood
-  = already selected candidate for
-  = already colored
-  = current reference

Region Growing as the first step for Segmentation?



# Region Growing with 'constant' seed point

Based on 8-adjacency, apply region growing on the following image considering:

(i) seed point (= seed pixel) = 6 (static seed, hence won't change in any iteration) and (ii) delta threshold intensity (between seed and neighbor)  $\leq 3$

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 5 | 6 | 6 | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6 | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4 | 4 | 3 | 2 | 5 | 6 |
| 5 | 4 | 5 | 4 | 2 | 3 | 4 | 6 |
| 0 | 3 | 2 | 3 | 3 | 2 | 4 | 7 |
| 1 | 0 | 0 | 0 | 2 | 2 | 5 | 6 |
| 1 | 1 | 1 | 1 | 0 | 3 | 4 | 4 |
| 1 | 0 | 1 | 0 | 2 | 3 | 5 | 4 |

# Region Growing with 'constant' seed point

Based on 8-adjacency, apply region growing on the following image considering:

(i) seed point (= seed pixel) = 6 (**constraint: ref doesn't roll over. It is always 6**) and (ii) threshold (= delta pixel intensity)  $\leq 3$

Lets identify the seeds

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 5 | 6 | 6 | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6 | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4 | 4 | 3 | 2 | 5 | 6 |
| 5 | 4 | 5 | 4 | 2 | 3 | 4 | 6 |
| 0 | 3 | 2 | 3 | 3 | 2 | 4 | 7 |
| 1 | 0 | 0 | 0 | 2 | 2 | 5 | 6 |
| 1 | 1 | 1 | 1 | 0 | 3 | 4 | 4 |
| 1 | 0 | 1 | 0 | 2 | 3 | 5 | 4 |

# Region Growing with 'constant' seed point

Based on 8-adjacency, apply region growing on the following image considering:  
(i) seed point (= seed pixel) = 6 and (ii) threshold  $\leq 3$

let's take one seed at a  
time as ref

|   |    |   |   |   |   |   |   |
|---|----|---|---|---|---|---|---|
| 5 | 6★ | 6 | 7 | 6 | 7 | 6 | 6 |
| 6 | 7  | 6 | 7 | 5 | 5 | 4 | 7 |
| 6 | 6  | 4 | 4 | 3 | 2 | 5 | 6 |
| 5 | 4  | 5 | 4 | 2 | 3 | 4 | 6 |
| 0 | 3  | 2 | 3 | 3 | 2 | 4 | 7 |
| 1 | 0  | 0 | 0 | 2 | 2 | 5 | 6 |
| 1 | 1  | 1 | 1 | 0 | 3 | 4 | 4 |
| 1 | 0  | 1 | 0 | 2 | 3 | 5 | 4 |

# Region Growing with 'constant' seed point

Based on 8-adjacency, apply region growing on the following image considering:

(i) seed point (= seed pixel) = 6 only. (ref will always be 6, ref does not roll over) and (ii) threshold  $\leq 3$

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 5 | 6 | 6 | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6 | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4 | 4 | 3 | 2 | 5 | 6 |
| 5 | 4 | 5 | 4 | 2 | 3 | 4 | 6 |
| 0 | 3 | 2 | 3 | 3 | 2 | 4 | 7 |
| 1 | 0 | 0 | 0 | 2 | 2 | 5 | 6 |
| 1 | 1 | 1 | 1 | 0 | 3 | 4 | 4 |
| 1 | 0 | 1 | 0 | 2 | 3 | 5 | 4 |

5,7 will be part of the grown region since they meet the threshold criteria, but they could not be considered as next new seed and initiate growing from them as starting point

# Region Growing with 'constant' seed point

Based on 8-adjacency, apply region growing on the following image considering:  
(i) seed point (= seed pixel) = 6 and (ii) threshold  $\leq 3$

|   |   |    |   |   |   |   |   |
|---|---|----|---|---|---|---|---|
| 5 | 6 | 6★ | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6  | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4  | 4 | 3 | 2 | 5 | 6 |
| 5 | 4 | 5  | 4 | 2 | 3 | 4 | 6 |
| 0 | 3 | 2  | 3 | 3 | 2 | 4 | 7 |
| 1 | 0 | 0  | 0 | 2 | 2 | 5 | 6 |
| 1 | 1 | 1  | 1 | 0 | 3 | 4 | 4 |
| 1 | 0 | 1  | 0 | 2 | 3 | 5 | 4 |

|   |   |    |   |   |   |   |   |
|---|---|----|---|---|---|---|---|
| 5 | 6 | 6★ | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6  | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4  | 4 | 3 | 2 | 5 | 6 |
| 5 | 4 | 5  | 4 | 2 | 3 | 4 | 6 |
| 0 | 3 | 2  | 3 | 3 | 2 | 4 | 7 |
| 1 | 0 | 0  | 0 | 2 | 2 | 5 | 6 |
| 1 | 1 | 1  | 1 | 0 | 3 | 4 | 4 |
| 1 | 0 | 1  | 0 | 2 | 3 | 5 | 4 |

Please complete it!

# Region Growing with seed point:

Based on 8-adjacency, apply region growing on the following image considering:  
(i) seed point (= seed pixel) = 6 and (ii) threshold  $\leq 3$

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 5 | 6 | 6 | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6 | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4 | 4 | 3 | 2 | 5 | 6 |
| 5 | 4 | 5 | 4 | 2 | 3 | 4 | 6 |
| 0 | 3 | 2 | 3 | 3 | 2 | 4 | 7 |
| 1 | 0 | 0 | 0 | 2 | 2 | 5 | 6 |
| 1 | 1 | 1 | 1 | 0 | 3 | 4 | 4 |
| 1 | 0 | 1 | 0 | 2 | 3 | 5 | 4 |

# Region Growing with seed point

Based on 8-adjacency, apply region growing on the following image considering:  
(i) seed point (= seed pixel) = 6 and (ii) threshold  $\leq 3$

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 5 | 6 | 6 | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6 | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4 | 4 | 3 | 2 | 5 | 6 |
| 5 | 4 | 5 | 4 | 2 | 3 | 4 | 6 |
| 0 | 3 | 2 | 3 | 3 | 2 | 4 | 7 |
| 1 | 0 | 0 | 0 | 2 | 2 | 5 | 6 |
| 1 | 1 | 1 | 1 | 0 | 3 | 4 | 4 |
| 1 | 0 | 1 | 0 | 2 | 3 | 5 | 4 |

# What next: Segmented output image

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 5 | 6 | 6 | 7 | 6 | 7 | 6 | 6 |
| 6 | 7 | 6 | 7 | 5 | 5 | 4 | 7 |
| 6 | 6 | 4 | 4 | 0 | 0 | 5 | 6 |
| 5 | 4 | 5 | 0 | 0 | 0 | 4 | 6 |
| 0 | 0 | 0 | 0 | 0 | 0 | 4 | 7 |
| 0 | 0 | 0 | 0 | 0 | 0 | 5 | 6 |
| 0 | 0 | 0 | 0 | 0 | 0 | 4 | 4 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

i. Desired part to be segmented

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

ii. Mask creation for further segmentation possible too (ex. lung separation from 100 images of same size)



# Outline

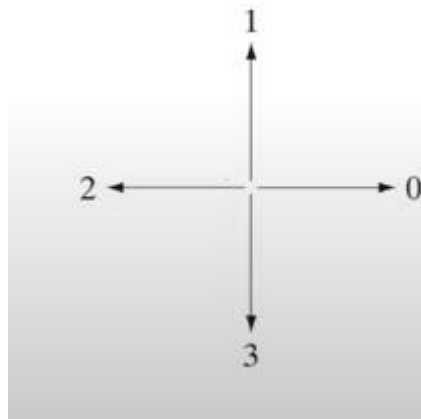
## Binary Image Processing

- Morphological Operations
- Labeling Regions
- Computing Distance in a Digital Image
- Region Properties
- Boundary Representations

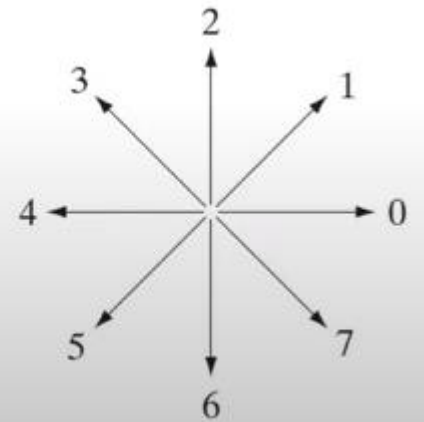
# Boundary Detection with Chain Code

The boundary detection problems can be easily solved by chain codes

Direction numbers



4-directional chain codes

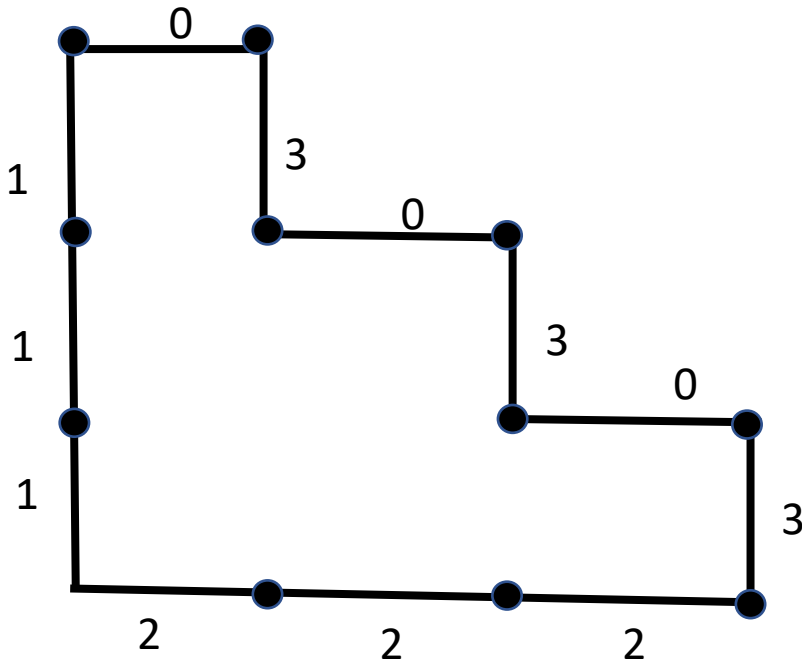


8-directional chain codes

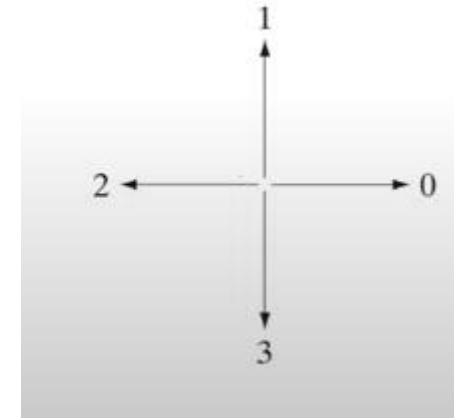
# Boundary Detection with Chain Code

The boundary detection problems can be easily solved by chain codes

Map the direction in image appropriately → join them in clockwise direction → determine pixel location of the boundary → segment out that



Chain-code for the boundary: 030303222111



4-directional chain codes